

This lecture explains the differences between instance methods, class methods, and static methods in object-oriented programming.

- Instance Methods:
 - An instance method is associated with an object (instance) of a class.
 - It operates on the data specific to that instance.
 - The `self` parameter refers to the instance itself, allowing access to its attributes.
 - Each instance can have its own version of the data, and the method's behavior depends on the instance it's called on.
- Class Methods:
 - A class method is bound to the class and not the instance of the class.
 - It receives the class itself as the first argument, conventionally named `cls`.
 - Class methods can access and modify class-level attributes (variables shared by all instances).
 - They are defined using the `@classmethod` decorator.
 - Class methods can be called either on the class itself or on an instance of the class, but the class is always passed as the implicit first argument.
- Static Methods:
 - A static method is not bound to either the instance or the class.
 - It's essentially a regular function that belongs to the class's namespace.
 - It doesn't receive any implicit first argument (`self` or `cls`).
 - Static methods are defined using the `@staticmethod` decorator.
 - They are called on the class itself (e.g., `ClassName.static_method()`).
 - Static methods are used when a method doesn't need access to instance-specific or class-specific data but logically belongs to the class.

Key Takeaways:

- Instance methods operate on instance-specific data.
- Class methods operate on class-level data and can modify the class state.
- Static methods are independent of both instances and the class, acting as utility functions within the class's namespace.
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